

Stage 4: Production API Specification & Swagger Documentation

Alight International Multi-Vendor Marketplace Document Version: 4.0.0 (Production-Ready) API Standard: RESTful JSON v1.0.0 (/api/v1/) OpenAPI Specification URL: GET /v3/api-docs Interactive Swagger UI: http://localhost:8080/swagger-ui.html

1. Global API Standards

Base URLs by Environment

Environment	Base URL	Protocol
Local Development	http://localhost:8080	HTTP
Staging	https://api-staging.alight.com	HTTPS
Production	https://api.alight.com	HTTPS (TLS 1.3)

Authentication Header

All authenticated endpoints require an HTTP Authorization header containing a valid JWT access token:

```
HTTP
Authorization: Bearer <JWT_ACCESS_TOKEN>
```

Standard Unified Response Envelope (ApiResponse<T>)

Every API endpoint returns a deterministic JSON envelope:

```
JSON
{
  "success": true,
  "message": "Operation completed successfully",
  "data": { ... },
  "error": null,
  "timestamp": "2026-09-12T07:45:00Z",
  "traceId": "c4b9-8e21-4f10-91ab",
  "path": "/api/v1/products"
}
```

Standard Error Response Envelope

```
JSON

{
  "success": false,
  "message": "Validation failed on 1 field",
  "data": null,
  "error": {
    "code": "VALIDATION_FAILED",
    "message": "Invalid SKU format",
    "status": 400,
    "details": [
      {
        "field": "sku",
        "message": "SKU must not be empty"
      }
    ]
  },
  "timestamp": "2026-09-12T07:45:00Z",
  "traceId": "c4b9-8e21-4f10-91ab",
  "path": "/api/v1/products"
}
```

2. Complete Endpoint Catalog by Domain Group

Group 1: Authentication & Session (/api/v1/auth/)

Method	Endpoint	Auth	Description	Status Codes
POST	/api/v1/auth/register	Public	Register new customer or vendor account	201, 400, 409
POST	/api/v1/auth/login	Public	Authenticate user credentials & issue JWT tokens	200, 401, 403
POST	/api/v1/auth/refresh-token	Public	Refresh expired JWT access token via rotated refresh token	200, 401
POST	/api/v1/auth/logout	Authenticated	Revoke active refresh tokens	200
GET	/api/v1/auth/me	Authenticated	Get current authenticated user profile	200, 401

Group 2: User Account & RBAC Permissions

Method	Endpoint	Auth	Description	Status Code
GET	/api/v1/users/profile	Authenticated	Retrieve customer personal profile	204
PUT	/api/v1/users/profile	Authenticated	Update personal profile details	204
GET	/api/v1/users/addresses	Authenticated	List user saved shipping & billing addresses	204
POST	/api/v1/users/addresses	Authenticated	Add a new shipping address	204
PUT	/api/v1/users/addresses/{id}/default	Authenticated	Set address as primary default	204
DELETE	/api/v1/users/addresses/{id}	Authenticated	Delete saved shipping address	204
GET	/api/v1/users/me/permissions	Authenticated	Introspect active roles, 45 permissions, and vendor ID	204
GET	/api/v1/admin/rbac/roles	ROLE_ADMIN	List all 4 roles and their bound permissions	204
GET	/api/v1/admin/rbac/permissions	ROLE_ADMIN	List all 45 fine-grained system permissions	204
GET	/api/v1/admin/rbac/users/{userId}/roles	ROLE_ADMIN	View target user's active roles	204

Method	Endpoint	Auth	Description	St Co
			& effective permissions	
PUT	/api/v1/admin/rbac/users/{userId}/roles	ROLE_SUPER_ADMIN	Assign or modify user roles	2 4 4
PUT	/api/v1/admin/rbac/roles/{roleName}/permissions	ROLE_SUPER_ADMIN	Update permissions assigned to a role	2 4

Group 3: Product Catalog, Taxonomy & Search

Method	Endpoint	Auth	Description	Status Codes
GET	/api/v1/categories	Public	List hierarchical category tree	200
GET	/api/v1/categories/{slug}	Public	Get category details by slug	200 , 404
GET	/api/v1/brands	Public	List active manufacturer & designer brands	200
GET	/api/v1/products	Public	Search and filter catalog (keyword, category, price, rating)	200
GET	/api/v1/products/{slug}	Public	Get complete product detail with variants, images & attributes	200 , 404
GET	/api/v1/products/featured	Public	Get curated high-rating featured products	200
POST	/api/v1/vendor/products	ROLE_VENDOR	Create draft hardware product	201 , 400
PUT	/api/v1/vendor/products/{id}	ROLE_VENDOR (Tenant)	Update owned vendor product	200 , 403 , 404
PUT	/api/v1/admin/products/{id}/status	ROLE_ADMIN	Approve or reject vendor product submission	200 , 403
GET	/api/v1/search/autocomplete	Public	Sub-10ms trigram autocomplete search suggestions	200
GET	/api/v1/bundles	Public	List curated hardware kits & modular package sets	200
POST	/api/v1/quotes	Public / Customer	Submit B2B architectural hardware RFQ with BOQ items	201 , 400

Group 4: Inventory & Warehouse Reservations

Method	Endpoint	Auth	Description	Status Codes
GET	/api/v1/inventory/products/{productId}/stock	Public	Check available stock across all warehouses	200 , 404
POST	/api/v1/inventory/reservations	Public / Customer	Create temporary checkout stock reservation lock (TTL 15m)	201 , 409
DELETE	/api/v1/inventory/reservations/{reservationId}	Public / Customer	Release reserved stock on cart cancellation	200
GET	/api/v1/vendor/warehouses	ROLE_VENDOR	List fulfillment warehouses and stock allocations	200
POST	/api/v1/vendor/inventory/adjust	ROLE_VENDOR	Reconcile stock counts & safety stock thresholds	200 , 400

Group 5: Shopping Cart, Multi-Vendor Checkout & Orders

Method	Endpoint	Auth	Description	Status Codes
GET	/api/v1/cart	Public / Session	Fetch active multi-vendor cart with subtotal & tax preview	200
POST	/api/v1/cart/items	Public / Session	Add item or variant to shopping cart	200 , 400 , 409
PUT	/api/v1/cart/items/{itemId}	Public / Session	Update item quantity in cart	200 , 404
DELETE	/api/v1/cart/items/{itemId}	Public / Session	Remove line item from cart	200 , 404
POST	/api/v1/checkout/process	Authenticated	Create multi-vendor split order (#10001 → #10001-A , #10001-B)	201 , 400
GET	/api/v1/orders	Authenticated	List customer personal order history	200
GET	/api/v1/orders/{orderNumber}	Authenticated (Owner)	Get master order summary & vendor breakdown	200 , 404
GET	/api/v1/vendor/orders	ROLE_VENDOR	List assigned vendor fulfillment sub-orders	200
PUT	/api/v1/vendor/orders/{subOrderNumber}/status	ROLE_VENDOR (Tenant)	Update sub-order fulfillment (PROCESSING, SHIPPED, DELIVERED)	200 , 403

Group 6: Payments, Webhooks & Settlements

Method	Endpoint	Auth	Description	Status Codes
POST	/api/v1/payments/create-intent	Authenticated	Initialize Razorpay / Stripe payment transaction intent	200 , 400
POST	/api/v1/payments/verify	Authenticated	Verify cryptographic payment signature	200 , 400
POST	/api/v1/payments/webhook	Public (Signature Verified)	Idempotent gateway webhook callback	200
GET	/api/v1/vendor/wallet	ROLE_VENDOR (Tenant)	Get vendor wallet balance, pending escrow & earnings	200
POST	/api/v1/vendor/wallet/payouts	ROLE_VENDOR	Request payout withdrawal to verified bank account	201 , 400
GET	/api/v1/admin/settlements/cycles	ROLE_ADMIN	View settlement cycles and payout batches	200 , 403
POST	/api/v1/admin/settlements/payouts/{id}/approve	ROLE_ADMIN	Approve vendor payout request	200 , 403

Group 7: Logistics, Returns, Reviews & Support

Method	Endpoint	Auth	Description	Status Codes
POST	/api/v1/logistics/calculate-rate	Public	Calculate shipping fee matrix by weight slab & pincode zone	200
GET	/api/v1/logistics/track/{trackingNumber}	Public	Real-time carrier checkpoint tracking history	200 , 404
POST	/api/v1/returns	Authenticated (Customer)	Submit 7-day return request with proof photos	201 , 400
PUT	/api/v1/vendor/returns/{id}/decision	ROLE_VENDOR (Tenant)	Accept or reject vendor return request	200 , 403
GET	/api/v1/reviews/product/{productId}	Public	List verified purchase reviews & helpful counts	200
POST	/api/v1/reviews	Authenticated (Customer)	Post verified product review with 1-5 rating & photos	201 , 400
POST	/api/v1/support/tickets	Authenticated	Create customer or vendor support dispute ticket	201 , 400
POST	/api/v1/support/tickets/{id}/messages	Authenticated (Owner)	Post reply message in support dispute thread	201 , 403

Group 8: System Observability & Audit

Method	Endpoint	Auth	Description	Status Codes
GET	/api/v1/health	Public	Live database connectivity and memory diagnostic probe	200 , 503
GET	/actuator/health	Public	Spring Boot Actuator liveness & readiness probes	200
GET	/api/v1/admin/audit-logs	ROLE_ADMIN	Inspect tamper-evident audit trail logs	200 , 403

3. Swagger UI Interactive Groups

The Swagger UI on `/swagger-ui.html` provides 6 pre-configured drop-down exploration groups:

1. `01-All-Marketplace-APIs` : Complete OpenAPI document encompassing all endpoints.
2. `02-Storefront-Public-APIs` : Consumer shopping experience, instant search, and reviews.
3. `03-Customer-Account-APIs` : Auth, profile, cart, checkout, payments, and personal orders.
4. `04-Vendor-Portal-APIs` : Seller dashboard, catalog creation, sub-order dispatch, and wallets.
5. `05-Admin-Operations-APIs` : Platform moderation, KYC approvals, and RBAC governance.
6. `06-System-Diagnostics-APIs` : Currencies, taxes, pricing matrix, and audit logs.